Dhinesh Rajasekaran

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Education

Master of Engineering - Robotics — Aug. 2022 - May. 2024
University of Maryland, College Park (3.7/4.0 GPA) — *College Park, MD, USA*

Bachelor of Technology - Electrical Engineering — Jul. 2017 - Jun. 2021
SRM Institute of Science and Technology, Chennai (8.8/10.0 GPA) — *Chennai, TN, India*

Technical Skills

Areas of Expertise	**High Speed Circuit Design**, Circuit Theory, **Flexible Electronics**
Tools/Technologies	**Altium**, **OrCAD**, SOLIDWORKS, Board Bring-up
Hardware Platforms	**AVR**, **ARM**, GPS, GSM, **STM, ESP**, **TI** MSP Boards & Sensors
Key Components	**LDO**, SMPS, Filters, ADC, DAC, **MOSFET**, Level Shifter
Instrumentation Platforms	Logic Analyzer, **Oscilloscope**, Spectrum Analyzer
Programming Languages	**Python**, C++/C
Proficient in Interfaces	I2C, SPI, UART, RS422, Ethernet, CSI, SMBus

Career Highlights

- **Electrical Engineer** with **3 years** of experience, **2 patents**, and expertise in **Schematics**, Component Selection, **Digital/Analog** Design, **PCB Layout**, **Sensors** Integration and Circuit Debugging.
- Demonstrated **technical expertise**, **attention to detail**, diligence, and **problem-solving** through successful execution of multiple long-term projects, professional experience and research initiatives.

Professional Experience

Bioptics Technology, Lexington — Oct. 2024 - Present
Hardware Engineer — *Lexington, KY, USA*

- Designed a **wearable** medical device for **fluorescence-guided brain surgery**, integrating a **NXP SoM**, high-power UV illumination, real-time **camera vision**, power electronics and **on-board data** processing for enhanced surgical precision.
- Designed a custom **NXP SoM**-based SBC with CSI, HDMI & USB interfaces and **multi-layer control PCBs** for **robotic systems**, integrating **motors**, LVDT and load cells with **IEC-compliant** hardware design.
- Developed a **closed-loop constant-current driver** for precise laser diode **current & optical power** regulation, utilizing input & output feedback monitoring to achieve **mV-level stability**, and ensuring protection against **overcurrent and thermal drift**.
- Developed a **LED illumination system** with efficient **thermal management** using a custom **MCPCB** and a novel rapid prototyping material using **PAHT-CF**.

Solinas Integrity, IIT Madras Research Park — Aug. 2021 - Jul. 2022
Robotics Hardware Engineer — *Chennai, TN, India*

- Engineered the **Electronic Architecture** and designed PCBs for a tethered **pipeline inspection robot** capable of detecting leaks, **corrosion**, and defects in underground water-filled pipelines.
- Led component selection and designed a **240W high-power transmission line**, associated **power electronics**, and **STM32**-based **high-speed digital circuitry** for a custom control PCB.
- Designed an **ATmega2560**-based **infotainment PCB** for the robot control station, integrating a joystick, TFT LCD, encoder mode selection, and a high-power connector interface.

Digital Blanket, Bangalore — Sept. 2021 - Jul. 2022
Embedded Hardware Engineer — *Bangalore, KA, India*

- Designed a **wet floor detection sensor** using a **FLIR** thermal camera and **ESP32**, implementing on-board image processing for precise hazard detection.
- Assisted in PCB layout for a **mmWave TI radar** sensor and developed a custom **smart home wireless sensor platform**, integrating **industrial-grade** sensors for **IAQ**, TVOC, light, and temperature monitoring.
- Developed novel sensor PCBs for **odor** detection and **human presence sensing**, including schematic and PCB design for the **Omron D6T MEMS** thermal sensor, with integration into **industrial automation** and IoT platforms.

Projects

Black Box Device for Marine Vessels:

- Designed a **tamper-proof, solar-powered** black box for ocean vessels, enabling data logging to a coast guard **dashboard**.
- Features include Tamper detection, **Iridium satellite** communication, and **SOS** broadcast buttons.

Bone Conduction & Accident Prevention Smart Helmet:

- Developed a **patented** project integrating **ADAS** into bike helmets for real-time safety alerts and infotainment features.
- Integrated **HUD** and bone conduction technology with a custom **Battery Management System** and **DAC circuit**.

Smart Kitchen Robot for Making Stuffed Indian Bread Variety:

- Developed the **world's first fully automated** cooking robot requiring only wheat and water inputs.
- Automated paratha making, **stacking**, and storing in **hotboxes** using **feedback circuits**, sensor PCB, and smart **IoT** control.

Custom Robotic Arm for Pick & Place Operations using Stereo Vision:

- Designed a 6-DoF manipulator with a 3D-printed design and custom **servo control PCB** for pick-and-place tasks.
- Programmed using **Embedded C++**, and compared its **motor** performance against the **UR5e** robotic arm.

Patents

HEAD GEAR SYSTEM AND METHOD FOR ENSURING THE SAFETY OF A RIDER OF A VEHICLE Dec. 2021
Patent No: 202141060755 Patent

- Patent published for the project "Bone Conduction & Accident Prevention Smart Helmet".

BAKER BOT SYSTEM, SMART KITCHEN ROBOT MACHINE, AND METHOD FOR AUTOMATIC MAKING OF CHAPATI Dec. 2021
Patent No: 202141060759 Patent

- Patent published for the project "Smart Kitchen Robot for Making Stuffed Indian Bread".